

1. Explain the challenges in Big Data

- Lack of Data Storage due to large scale of data increases every second
- Poor security due to poor data sharing mechanisms (cloud computing services, online purchases that leads to sharing of credit card details, etc
- Lack of data schema: Big data has no data schema
- Lack of data quality: Big data has poor/no data quality, it the job of data scientist to generate data quality with high accuracy in real-time manner

2. Explain the architectural layers in Big data

- Big Data Sources Layer: a big data environment can manage both batch processing and real-time processing of big data sources, such as data warehouses, relational database management systems, SaaS applications, and IoT devices.
- Storage Layer: receives data from the source, converts the data into a format comprehensible for the data analytics tool, and stores the data according to its format.
- Analysis Layer: analytics tools extract business intelligence from the big data storage layer.
- Consumption Layer: receives results from the big data analysis layer and presents them to the pertinent output layer - also known as the business intelligence layer.

3. Explain the data storage in Big data. E.g

.....lecture03.....

4. Differentiate between batch processing and real-time processing models

Feature	Batch Processing	Real-Time Processing
Definition	Data is collected, processed in bulk at scheduled intervals.	Data is processed instantly as it arrives.
Speed	Slower, as processing happens periodically.	Fast, as processing happens instantly.
Use Case	Suitable for large volumes of data that do not require immediate action.	Suitable for applications where instant response is required.
Computational Load	Can be optimized for efficiency but may require large resources at once.	Requires continuous processing and

.....lecture differences

	Batch Data Processing	Real-Time Data Processing
Hardware	Most storage and processing resources requirement to process large batches of data.	Less storage or no storage required to process the current data packets. Less computational requirements.
Performance	Latency could be minutes, hours, or days	Latency needs to be in seconds or milliseconds
Data set	Large batches of data	Few current data packet
Analysis	Complex computation and analysis of a larger time frame	Simple reporting or computation

5. Explain some examples of big data processing models

Some example in those case:

I. Batch processing model

- >Hadoop Mapreduce: example analysing large dataset(marketing analysis)
- >log processing ,eg website log traffic analys
- >Retail business , and wholesales analysis

II. Real time processing model

- >Real-time Sentiment Analysis
- >Real-time Weather Data Analysis
- >processing IoT sensor data for smart automation
- >recomandation engine
- >fraud detection in online banking

6. Explain the applications of Big Data analytic techniques

Big data analytics techniques refers to the methods used to examine data and generate useful insights. The following are some examples of Big data analytics techniques

➤ **Cluster analysis technique**

Application example:

- > market segmentation
- > Anomaly detection

➤ **Machine learning technique**

Application examples:

- > Social Media Features
- > Product Recommendations
- > Image Recognition
- > **Sentiment Analysis**

Application of Sentiment Analysis

- > Social media monitoring
- > Customer support management
- > Brand monitoring and reputation management

7. Explain the challenges of Big data analytic techniques

1. Data Volume

- **Challenge:** The sheer volume of data generated daily is enormous, making it difficult to store, process, and analyze.
- **Impact:** Traditional data storage and processing systems may not be capable of handling such large volumes, leading to the need for scalable solutions like distributed storage and computing frameworks (e.g., Hadoop, Spark).

2. Data Variety

- **Challenge:** Data comes in various formats, including structured (e.g., databases), semi-structured (e.g., XML, JSON), and unstructured (e.g., text, images, videos).

- **Impact:** Integrating and analyzing diverse data types require advanced tools and techniques, such as natural language processing (NLP) for text and computer vision for images.

3. Data Velocity

- **Challenge:** Data is generated at high speeds, especially from real-time sources like social media, IoT devices, and sensors.
- **Impact:** Real-time processing and analysis are necessary to derive timely insights, which can be challenging to achieve with traditional batch processing methods.

4. Data Veracity

- **Challenge:** Ensuring the accuracy, consistency, and reliability of data is difficult, especially when dealing with large datasets from multiple sources.
- **Impact:** Poor data quality can lead to incorrect analyses and misleading insights, necessitating robust data cleaning and validation processes.

5. Data Complexity

- **Challenge:** Data often comes from multiple sources and may have complex relationships and interdependencies.
- **Impact:** Analyzing such complex data requires sophisticated algorithms and models, which can be computationally intensive and difficult to implement.

6. Scalability

- **Challenge:** As data grows, the systems and algorithms used for analysis must scale accordingly.
- **Impact:** Ensuring scalability without compromising performance is a significant challenge, often requiring distributed computing and parallel processing techniques.

7. Privacy and Security

- **Challenge:** Protecting sensitive information and ensuring data privacy is crucial, especially with regulations like GDPR and CCPA.
- **Impact:** Implementing robust security measures and ensuring compliance with legal requirements can be complex and resource-intensive.

8. Skill Gap

- **Challenge:** There is a shortage of skilled professionals who are proficient in big data technologies and analytics techniques.
- **Impact:** Organizations may struggle to find and retain talent, leading to delays and inefficiencies in data analytics projects.

9. Cost

- **Challenge:** Storing, processing, and analyzing large datasets can be expensive, requiring significant investment in infrastructure and tools.
- **Impact:** Organizations need to balance the cost of big data initiatives with the potential return on investment (ROI).

10. Integration with Existing Systems

- **Challenge:** Integrating big data analytics tools and platforms with existing IT infrastructure and legacy systems can be difficult.
- **Impact:** Ensuring seamless integration requires careful planning and may involve significant customization and development efforts.

11. Ethical Considerations

- **Challenge:** The use of big data analytics raises ethical questions, such as the potential for bias in algorithms and the impact on individuals' privacy.
- **Impact:** Organizations must navigate these ethical considerations carefully to maintain public trust and avoid legal repercussions.

12. Real-time Analytics

- **Challenge:** Performing real-time analytics on streaming data is complex and requires specialized tools and techniques.
- **Impact:** Achieving low-latency processing and real-time insights can be challenging, especially with high-velocity data streams.

13. Data Governance

- **Challenge:** Establishing policies and procedures for data management, including data ownership, access controls, and audit trails, is essential but challenging.
- **Impact:** Effective data governance ensures data quality and compliance but requires ongoing effort and resources.

14. Algorithm Complexity

- **Challenge:** Developing and implementing complex algorithms for big data analytics can be difficult and time-consuming.
- **Impact:** Ensuring that algorithms are efficient, accurate, and scalable is a significant challenge, often requiring specialized expertise.

15. Interdisciplinary Collaboration

- **Challenge:** Big data analytics often requires collaboration between data scientists, domain experts, IT professionals, and business stakeholders.
- **Impact:** Effective communication and collaboration across different disciplines can be challenging but are essential for successful analytics projects.

Addressing these challenges requires a combination of advanced technologies, skilled personnel, and robust processes. Organizations must invest in the right tools and infrastructure, foster a data-driven culture, and continuously adapt to the evolving landscape of big data analytics.